

EEET202

CHINES & TRANSFORMERS

MODULE:2

Tutorial – 1

e performance characteristics of DC generators

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ator delivers 195 A at a terminal voltage of 250 V. The armature and shunt field resistances are $0.02\ \Omega$ and $50\ \Omega$ respectively. The iron and frictional losses are equal to 950 W. Find (i) the generated e.m.f. (ii) the mechanical power input (iii) the output of the motor (iv) commercial, mechanical and electrical efficiencies.

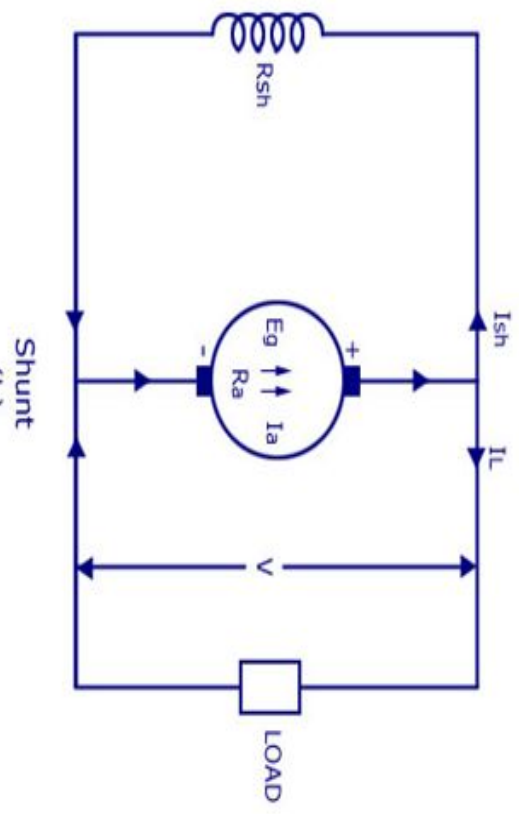
$$I_L = 195\text{ A}$$

$$V = 250\text{ V}$$

$$R_a = 0.02\ \Omega$$

$$R_{sh} = 50\ \Omega$$

$$W_c = 950\text{ W}$$



ator delivers 195 A at a terminal voltage of 250 V. The iron and shunt field resistances are $0.02\ \Omega$ and $50\ \Omega$. The iron and frictional losses are equal to 950 W. Find (i) the induced e.m.f. (ii) the mechanical power (iii) the output of the generator (iv) commercial, mechanical and electrical efficiency.

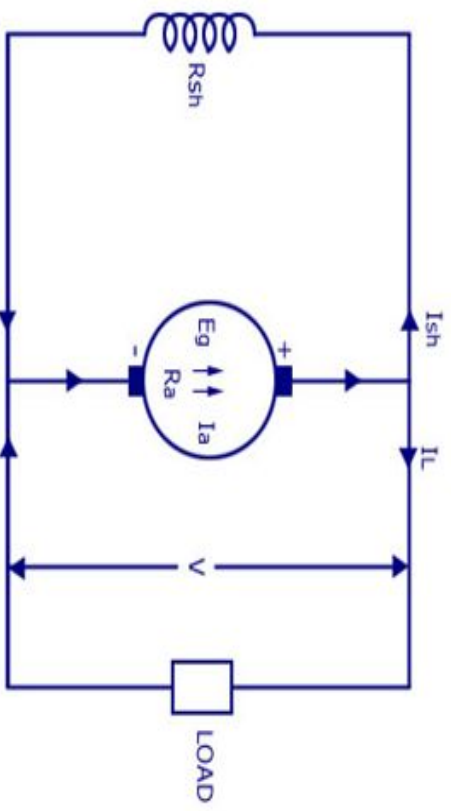
$$\text{Copper loss} = I_a^2 R_a$$

$$\text{Copper loss} = I_{sh}^2 R_{sh}$$

$$\text{Mechanical power} = P_{in}$$

$$P_{out} + \text{losses}$$

$$P_{out} + W_c + \text{Cu losses}$$



$$I_L = 195\ A \quad \text{Shunt}$$

$$V = 250\ V$$

$$R_a = 0.02\ \Omega$$

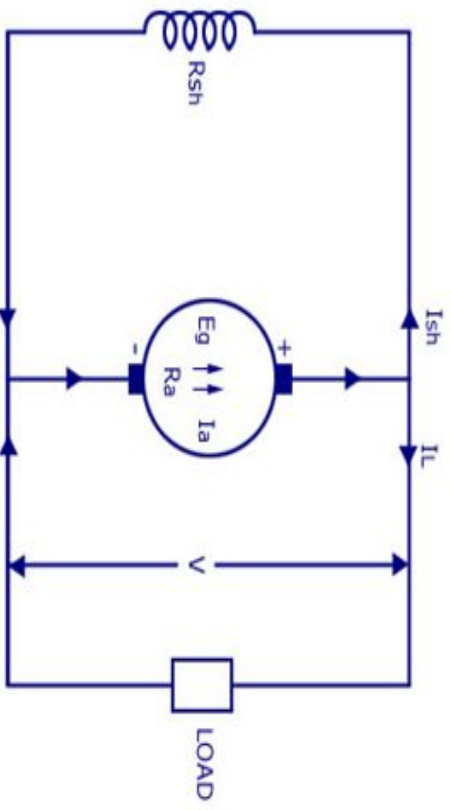
$$R_{sh} = 50\ \Omega$$

$$W_c = 950\ W$$

ator delivers 195 A at a terminal voltage of 250 V. The armature and shunt field resistances are $0.02\ \Omega$ and $50\ \Omega$ respectively. The iron and frictional losses are equal to 950 W. Calculate (i) the mechanical power input (ii) the mechanical power developed in the armature (iii) the output of the generator (iv) commercial, mechanical and electrical efficiency.

$$\text{Efficiency} = \frac{P_{out}}{P_{in}}$$

$$\text{Efficiency} = \frac{E_g I_a}{P_{in}}$$



$$\text{Efficiency} = \frac{P_{out}}{E_g I_a}$$

$$I_L = 195\ A \quad \text{Shunt}$$

$$V = 250\ V$$

$$R_a = 0.02\ \Omega$$

$$R_{sh} = 50\ \Omega$$

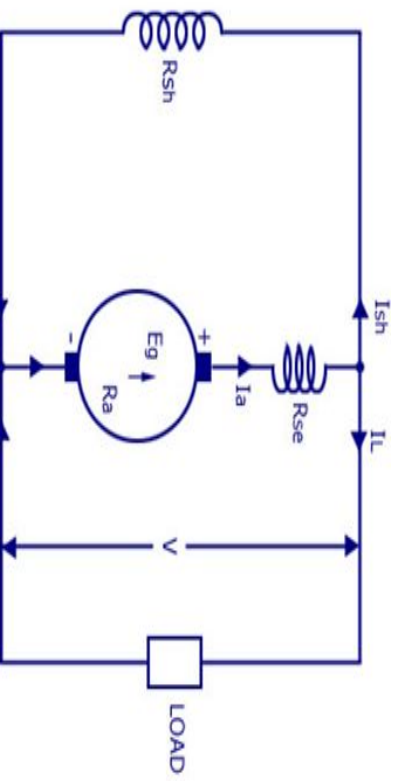
$$W_c = 950\ W$$

ator supplies 96 A at a terminal voltage of 200
are and field resistances are $0.1\ \Omega$ and $50\ \Omega$
The iron and frictional losses are 2500W.

ated

l efficiency

generator running at 1000 rpm delivers a load of terminal voltage of 220 V. The resistances of shunt field and series field are 0.04 Ω , 110 Ω and armature resistance is 0.05 Ω respectively. Overall efficiency at the above load is 85%. Calculate (i) iron and frictional loss (ii) copper loss (iii) torque of the prime mover



$$P_{out} = 20 \text{ KW} = 20000 \text{ W}$$

$$V = 220 \text{ V}$$

$$N = 1000 \text{ rpm}$$

$$R_a = 0.04 \Omega$$

$$R_{sh} = 110 \Omega$$

$$R_{se} = 0.05 \Omega$$

$$\eta = 85\% = 0.85$$

Iron losses

Armature reaction

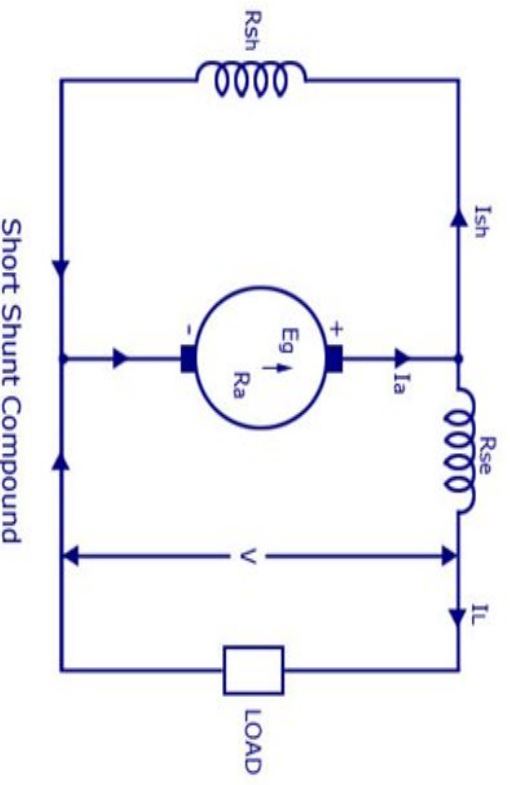
$$\omega = \frac{2\pi N}{60}$$

compound generator has armature, series field and resistances are $0.06\ \Omega$, $0.03\ \Omega$ and $110\ \Omega$ supplies 100 lamps rated at 250 V, 40 W. Find emf. Assume that contact drop per brush is 1 V.

$$V_b - I_a R_a - I_{se} R_{se}$$

$$= 40 \times 100$$

$$= V_L$$



$$V = 250V$$

$$R_a = 0.06\Omega$$

$$R_{se} = 0.03\Omega$$

$$R_{sh} = 110\Omega$$

$$V_b = 1V \text{ per brush}$$

$$\frac{I_{se} R_{se}}{R_{sh}}$$

$$+ I_{sh}$$